Segmentation crown tree

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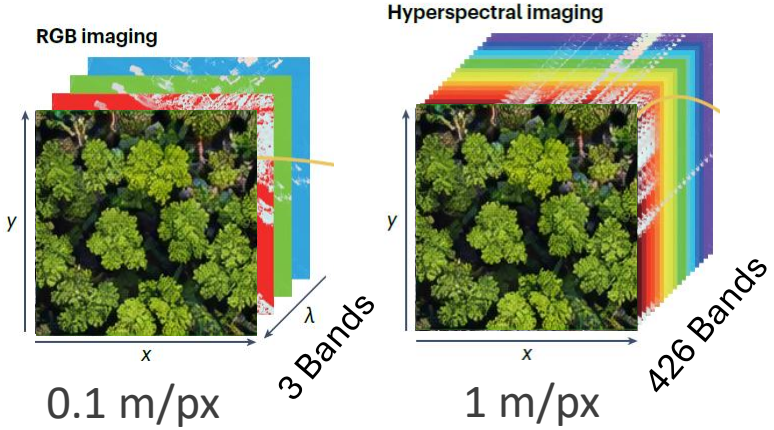
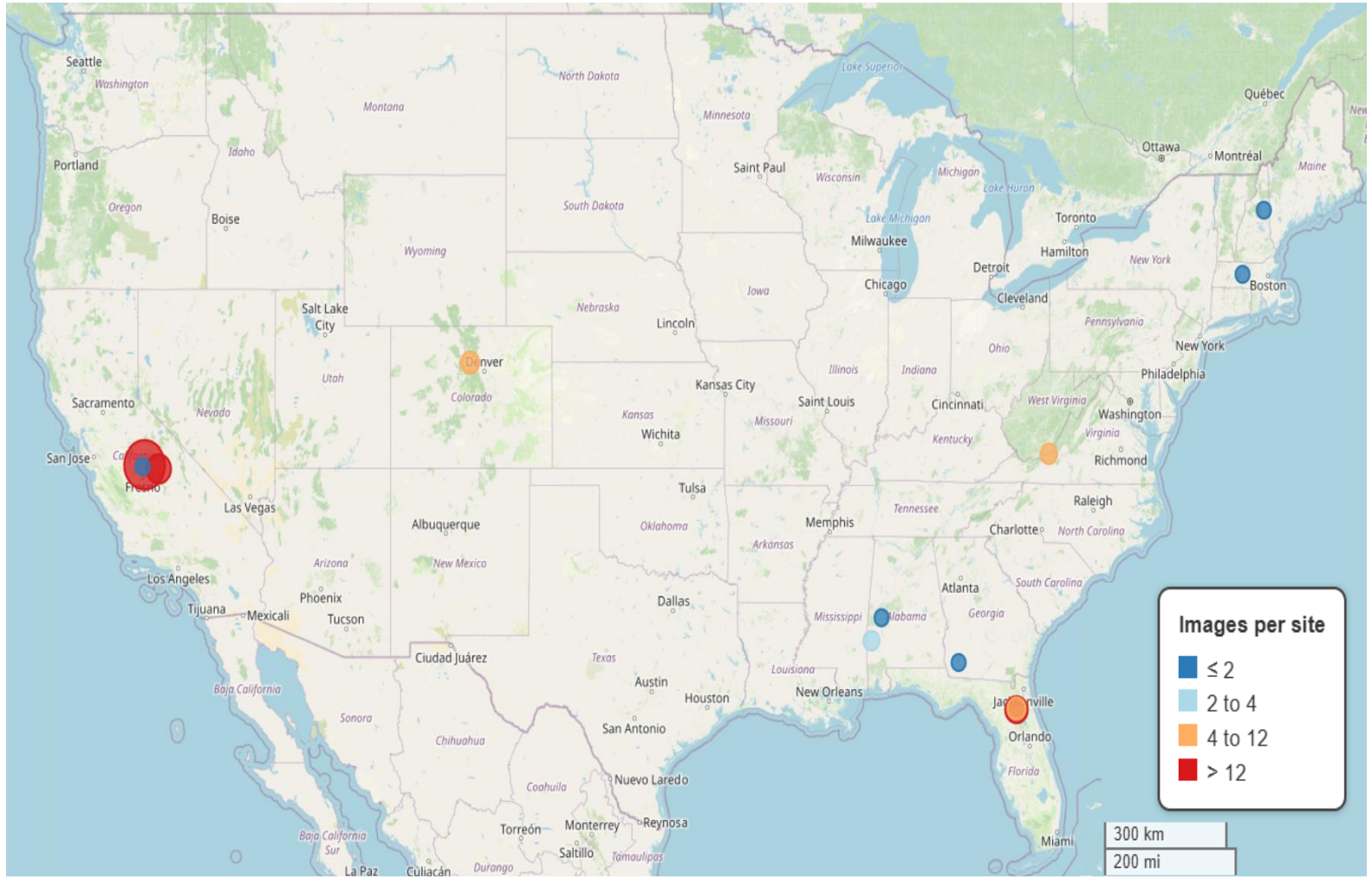
Motivation

- Forest inventory & biodiversity monitoring
- Carbon stock estimation
- Deforestation & Habitat Loss

Data description

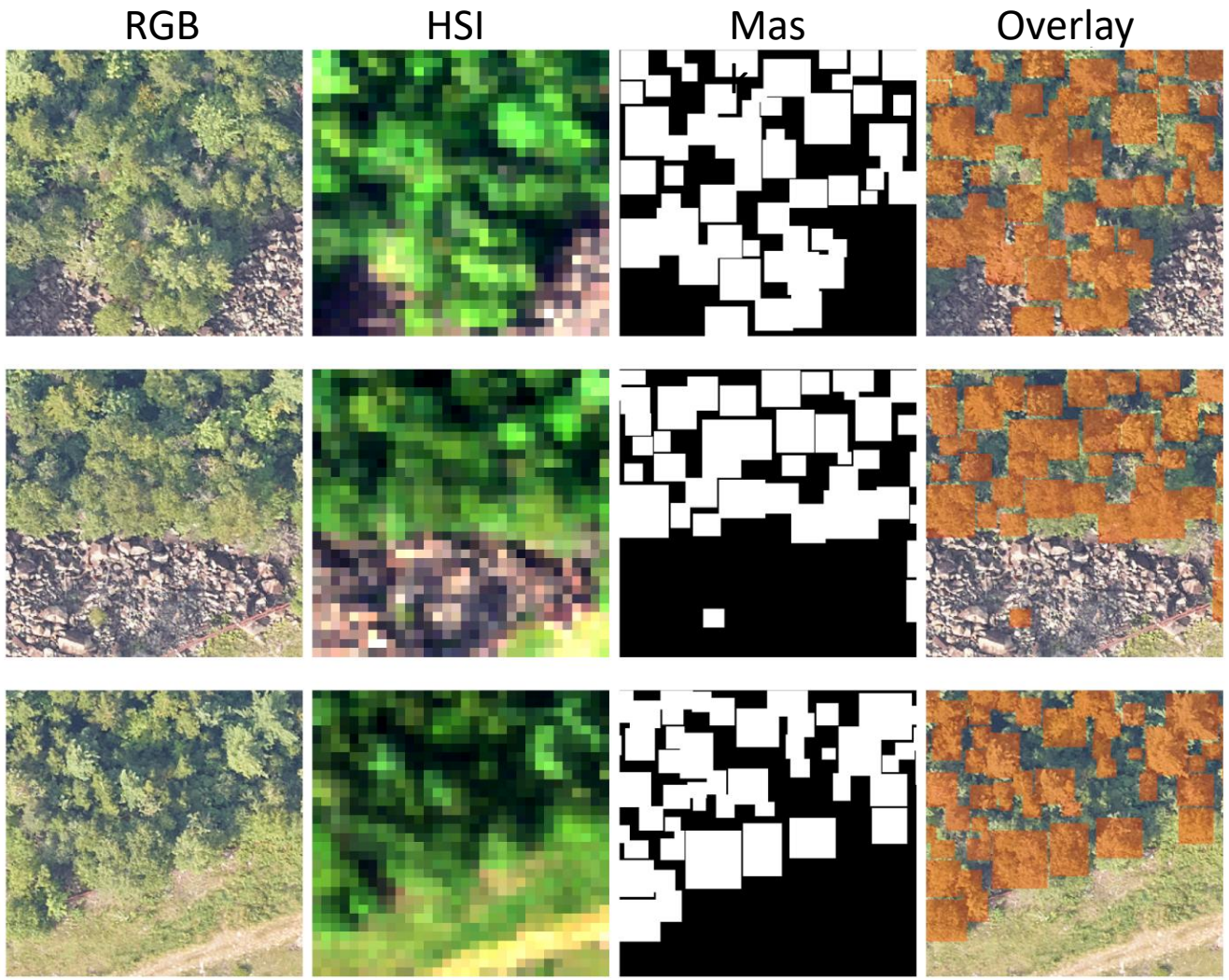
- 13 US NEON sites
- 16 RGB and HSI scenes
- Data recorded between 2018 and 2019





Data Processing Pipeline

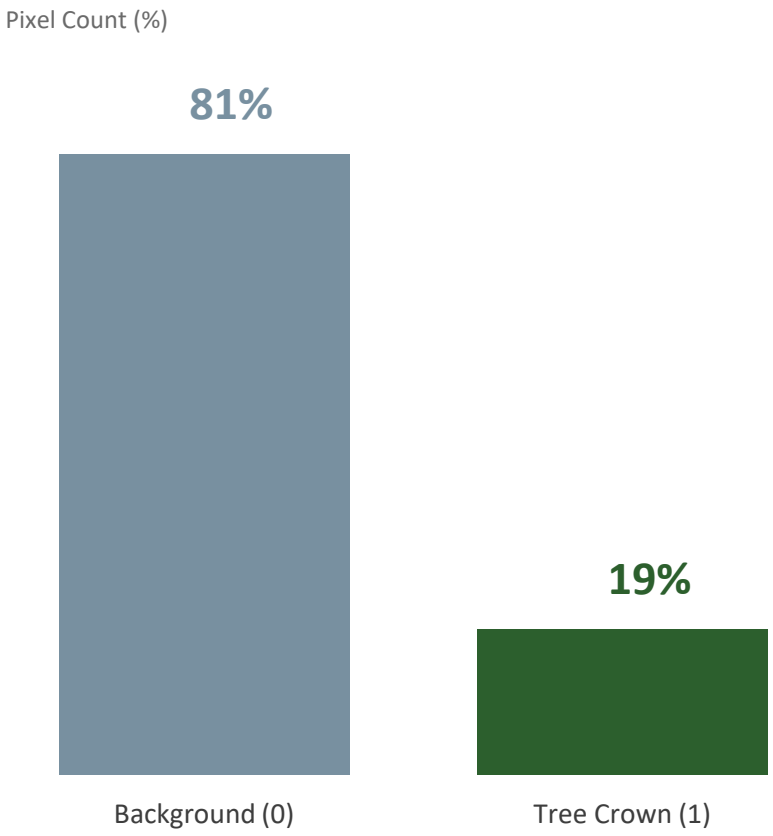
Generated by DALL·E



2202 patches of 320x320 RGB and 32x32 HSI

Loss design and metrics

Pixel Class Distribution



Dice Loss

$$L_{Dice} = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}$$

Where:
BCE: Binary cross entropy
z logits
p predicted probability
wp positive class weight
y ground truth mask
ϵ smoothing constant

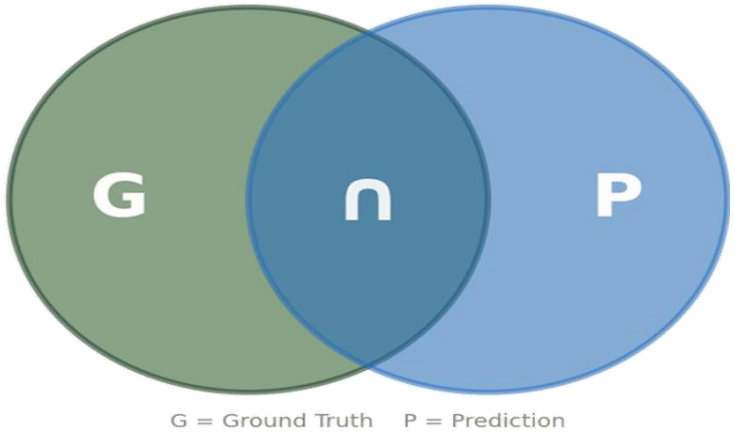
Focal Loss

$$L_{FL} = (1 - e^{-BCE})^{\gamma} \cdot BCE$$

$$BCE = -w_p y \log(\sigma(z)) - (1-y)\log(1-\sigma(z))$$

Total Loss

$$L = L_{FL} + \lambda L_{Dice}$$



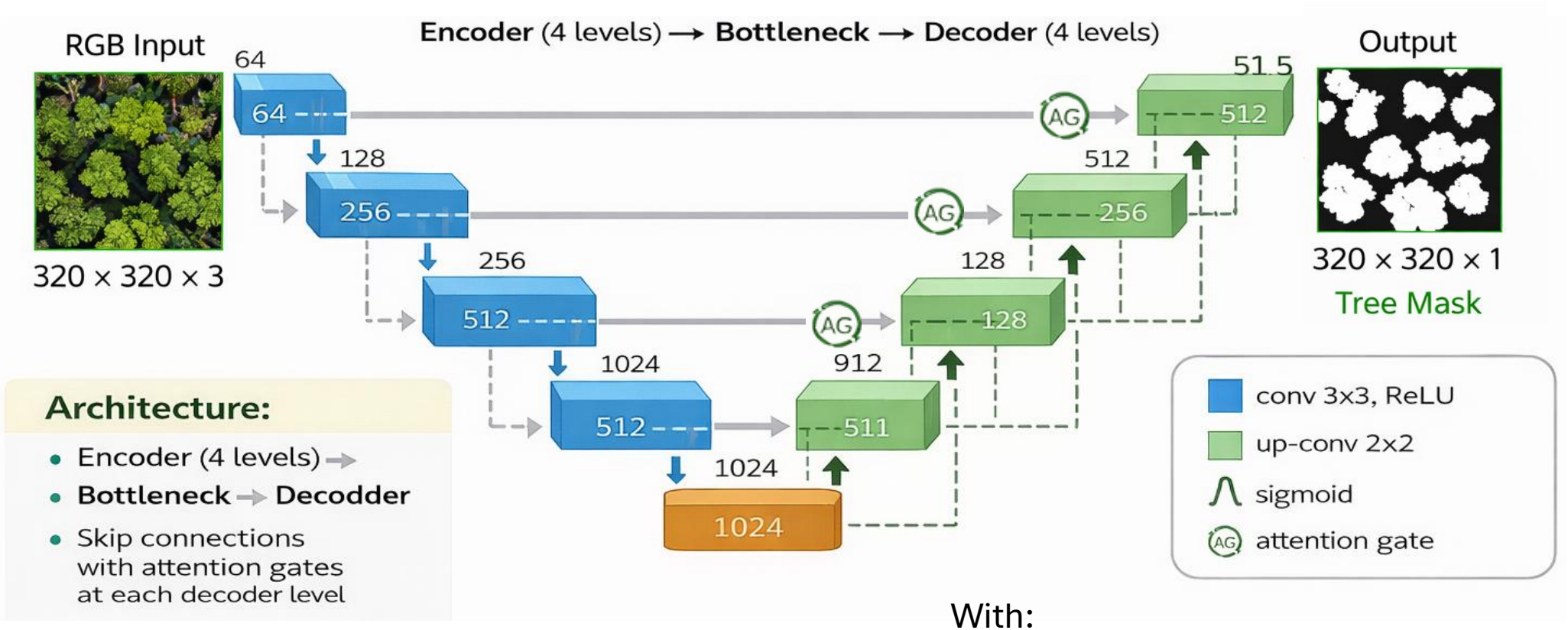
Dice score

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}$$

Intersection over union

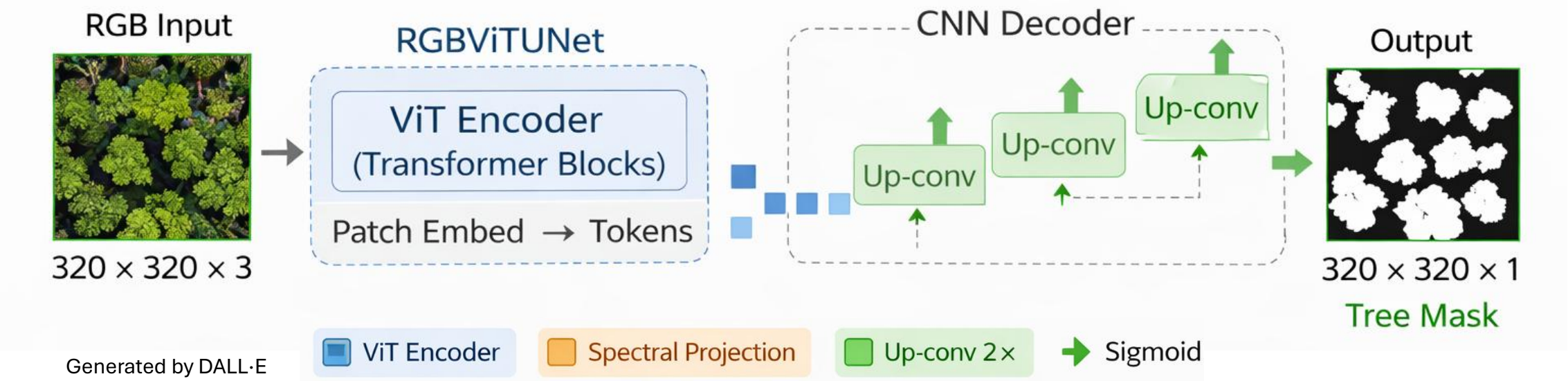
$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}$$

Models: Unet attention



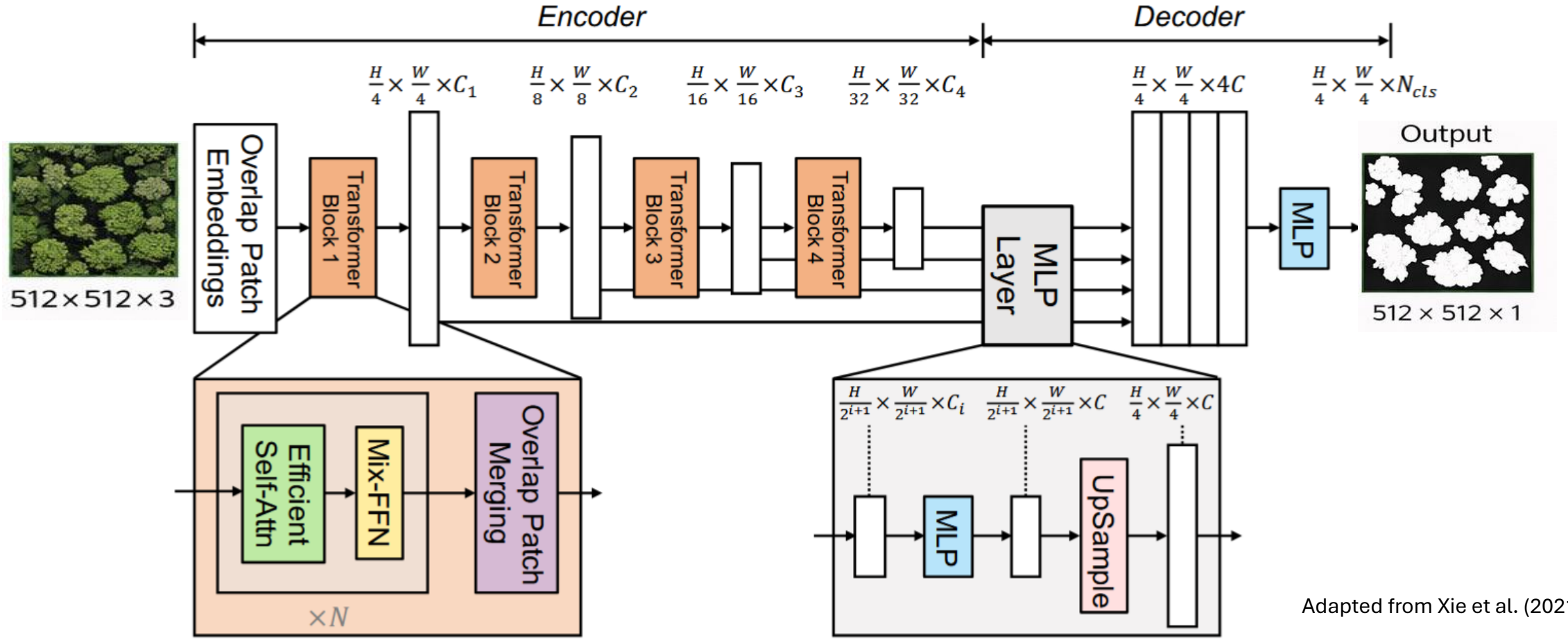
With:
- 2D CNN for RGB imagry
- 3D CNN for HSI imagry

Models: ViT Unet



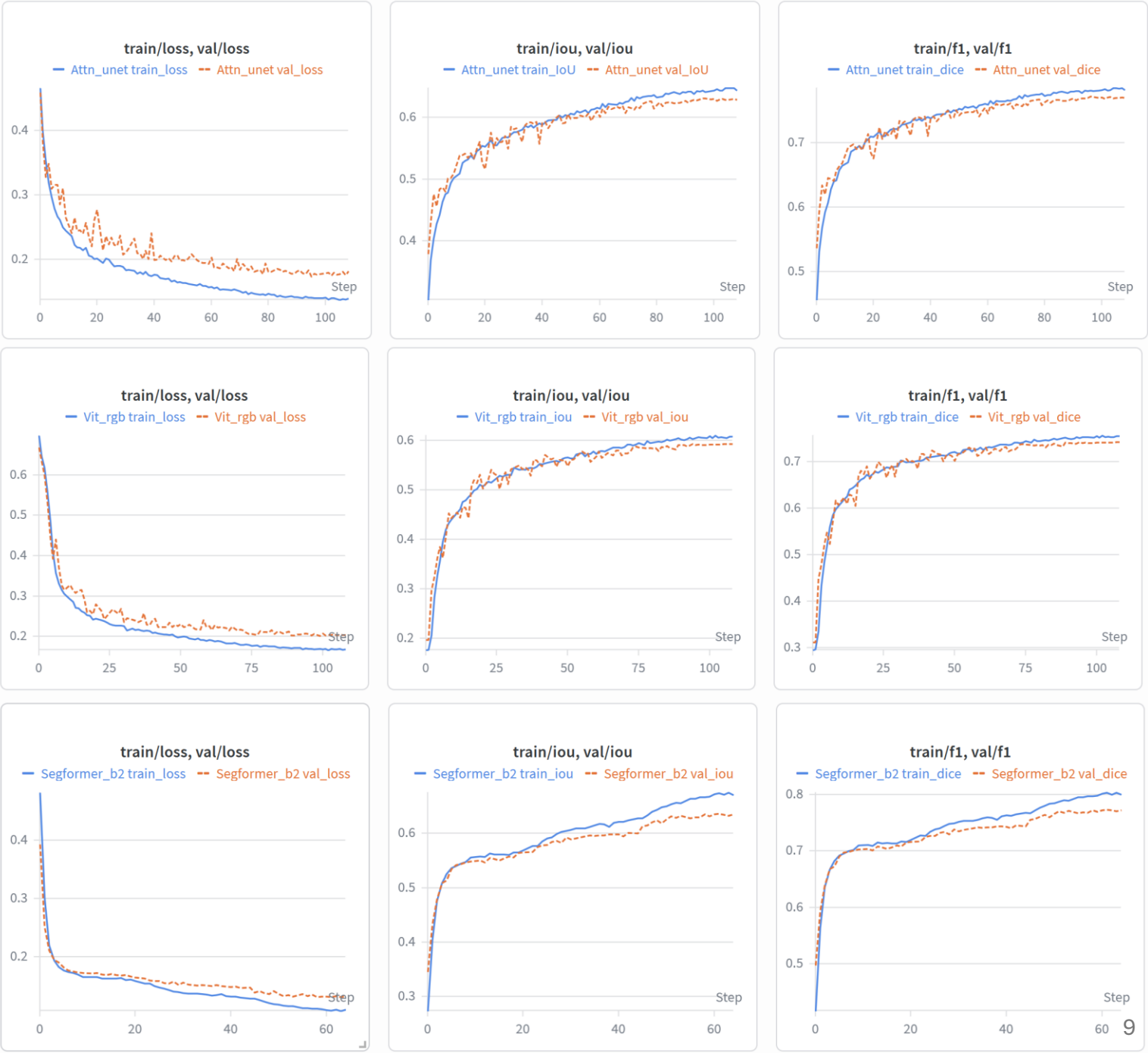
Generated by DALL·E

Models: SegFormer

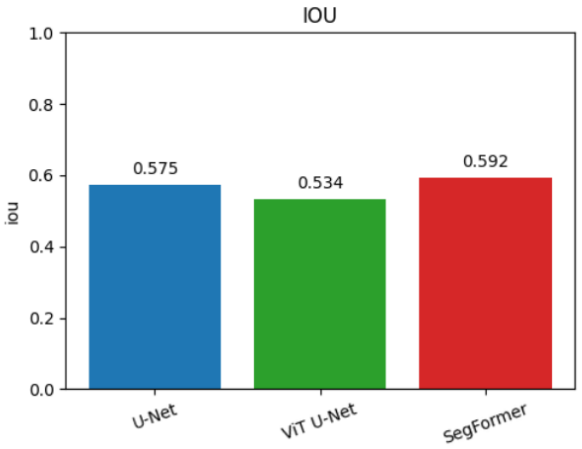


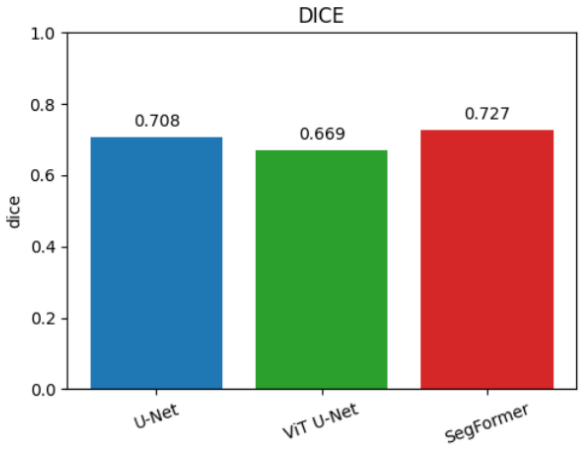
Adapted from Xie et al. (2021)

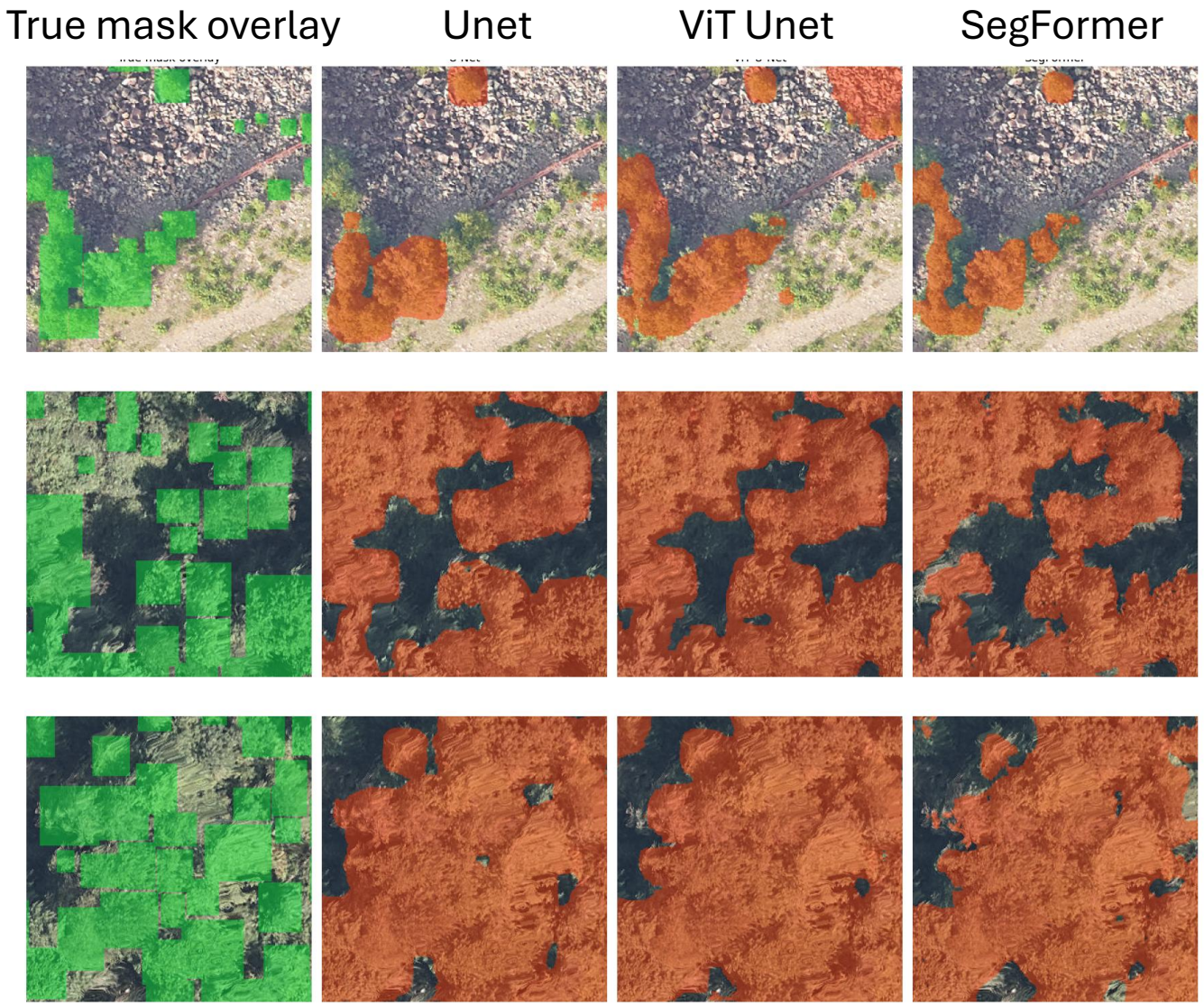
Results


train/loss, val/loss
Attn_unet train_loss
Attn_unet val_loss
train/iou, val/iou
Attn_unet train_IoU
Attn_unet val_IoU
train/f1, val/f1
Attn_unet train_dice
Attn_unet val_dice
train/loss, val/loss
Vit_rgb train_loss
Vit_rgb val_loss
train/iou, val/iou
Vit_rgb train_iou
Vit_rgb val_iou
train/f1, val/f1
Vit_rgb train_dice
Vit_rgb val_dice
train/loss, val/loss
Segformer_b2 train_loss
Segformer_b2 val_loss
train/iou, val/iou
Segformer_b2 train_iou
Segformer_b2 val_iou
train/f1, val/f1
Segformer_b2 train_dice
Segformer_b2 val_dice
Step

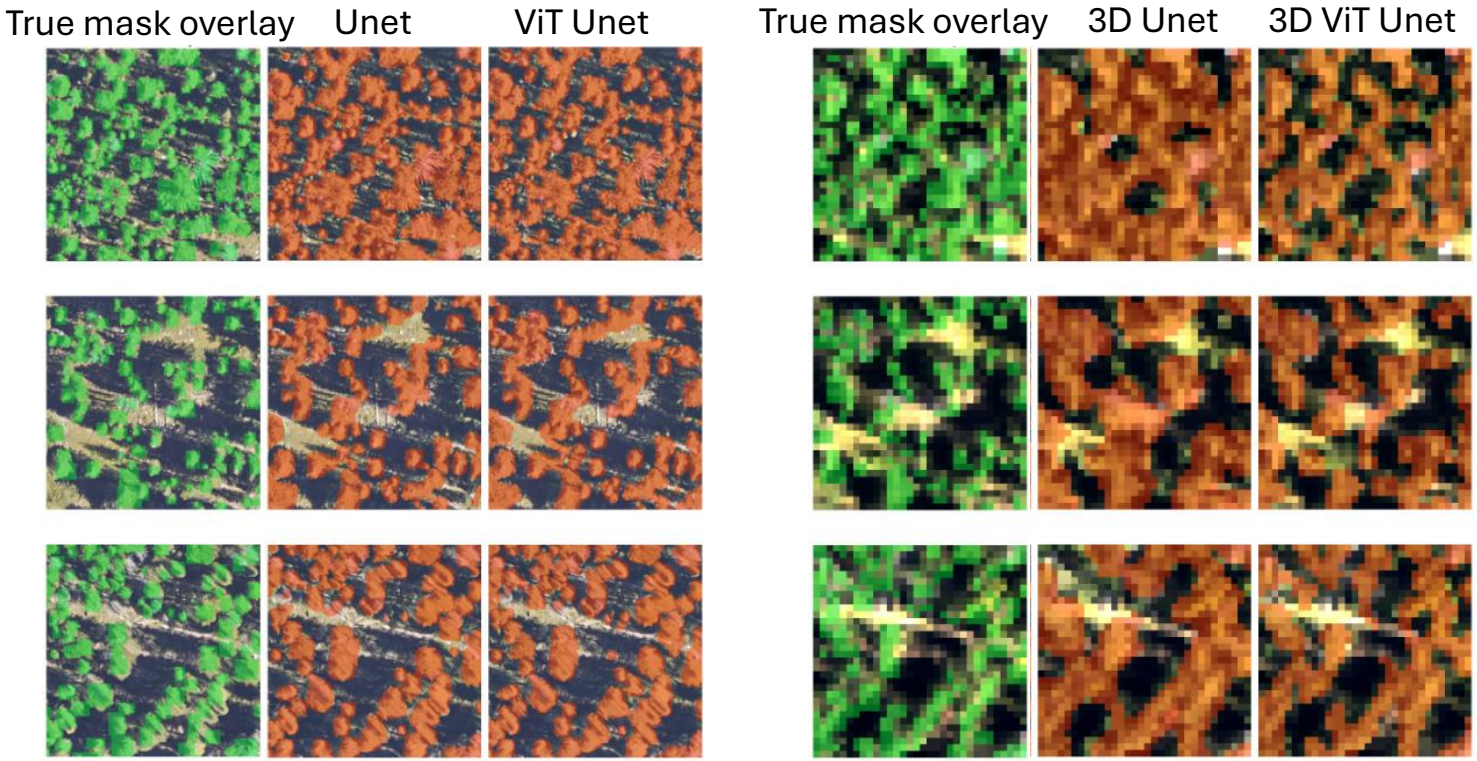
Results

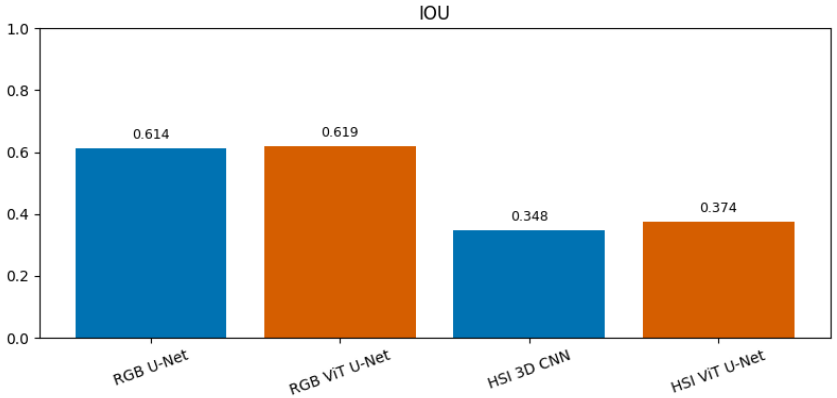

IOU
iou
1.0
0.8
0.6
0.4
0.2
0.0
0.575
0.534
0.592
U-Net
ViT U-Net
SegFormer

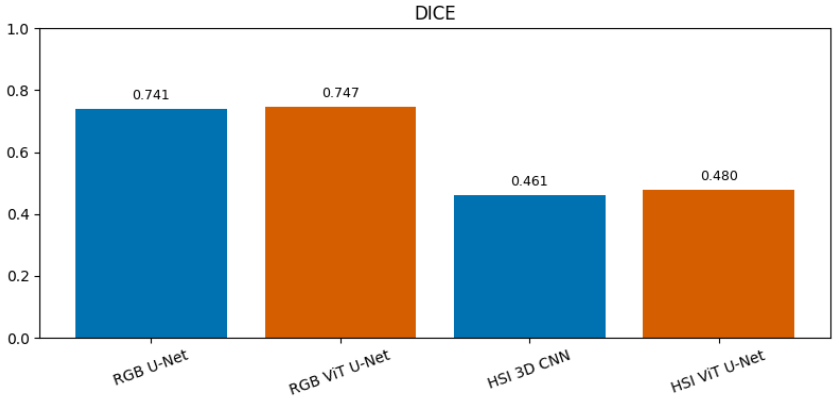

DICE
dice
1.0
0.8
0.6
0.4
0.2
0.0
0.708
0.669
0.727
U-Net
ViT U-Net
SegFormer


True mask overlay
Unet
ViT Unet
SegFormer

Results


True mask overlay
Unet
ViT Unet
True mask overlay
3D Unet
3D ViT Unet


IOU
1.0
0.8
0.6
0.4
0.2
0.0
0.614
0.619
0.348
0.374
RGB U-Net
RGB ViT U-Net
HSI 3D CNN
HSI ViT U-Net


DICE
1.0
0.8
0.6
0.4
0.2
0.0
0.741
0.747
0.461
0.480
RGB U-Net
RGB ViT U-Net
HSI 3D CNN
HSI ViT U-Net

Demo

🌲 Tree Crown Segmentation 🌲

Device: cuda

Upload RGB image (.tif/.tiff/.png/.jpg/.jpeg)

Drag and drop file here
Limit 200MB per file • TIF, TIFF, PNG, JPG, JPEG

Browse files

2018_SJER_3_252000_4108000_image_517.tif 0.5MB

Ecological report

Location (optional)

Center: 37.09315 N, -119.78202 E | CRS: EPSG:32611 | Bounds: [37.0930, -119.7823] -> [37.0933, -119.7818]

Year (optional)

2018

Compare with previous cover

Generate report

Here's the interpretation of the tree-crown segmentation scene:

Scene Summary The observed tree cover is 34.1%. The quantitative scene metrics indicate a uniform spatial distribution with an average crown area of 91.02 m^2, crown area variance of 16882.26 m^{22}, and a tree density of 37.5 trees/ha.

Ecological Interpretation Based on the observed canopy coverage and spatial distribution, it is likely that the canopy condition is relatively open with some structural heterogeneity. The uniform spatial distribution suggests that the vegetation is continuous, which may indicate a stable or regenerating ecosystem. However, without further information, it is difficult to determine if this is a degraded or healthy forest.

Biomass and Carbon Implications While we cannot directly measure biomass or carbon from RGB crown cover, the observed canopy coverage can provide some qualitative insights. A moderate canopy coverage of 34.1% may indicate a relatively stable ecosystem with some level of structural complexity. However, this interpretation is limited by the lack of information on tree height, DBH, and understory vegetation.

Biodiversity and Habitat Implications The uniform spatial distribution and moderate canopy coverage suggest that habitat availability is relatively consistent across the scene. Canopy connectivity appears to be high, which may facilitate species movement and dispersal. However, fragmentation index values indicate a low level of fragmentation, suggesting that the ecosystem may have some structural complexity.

Change Assessment Since there is no previous observation available, we cannot assess temporal change.

Limitations Key limitations of this RGB-only analysis include:

- Missing species identity
- Tree height and DBH information
- Understory vegetation data
- Need for field validation to confirm ecological interpretations

Recommendations

1. **Conduct a LiDAR survey**: To obtain more accurate information on tree height, DBH, and understory vegetation.
2. **Integrate multispectral or hyperspectral data**: To gain insights into species composition and canopy condition.
3. **Field validation**: To confirm ecological interpretations and collect additional data on tree species, density, and structure.

Please note that these recommendations are based on the limitations of the provided RGB-only analysis and may not be applicable to all scenarios.

Geospatial Map

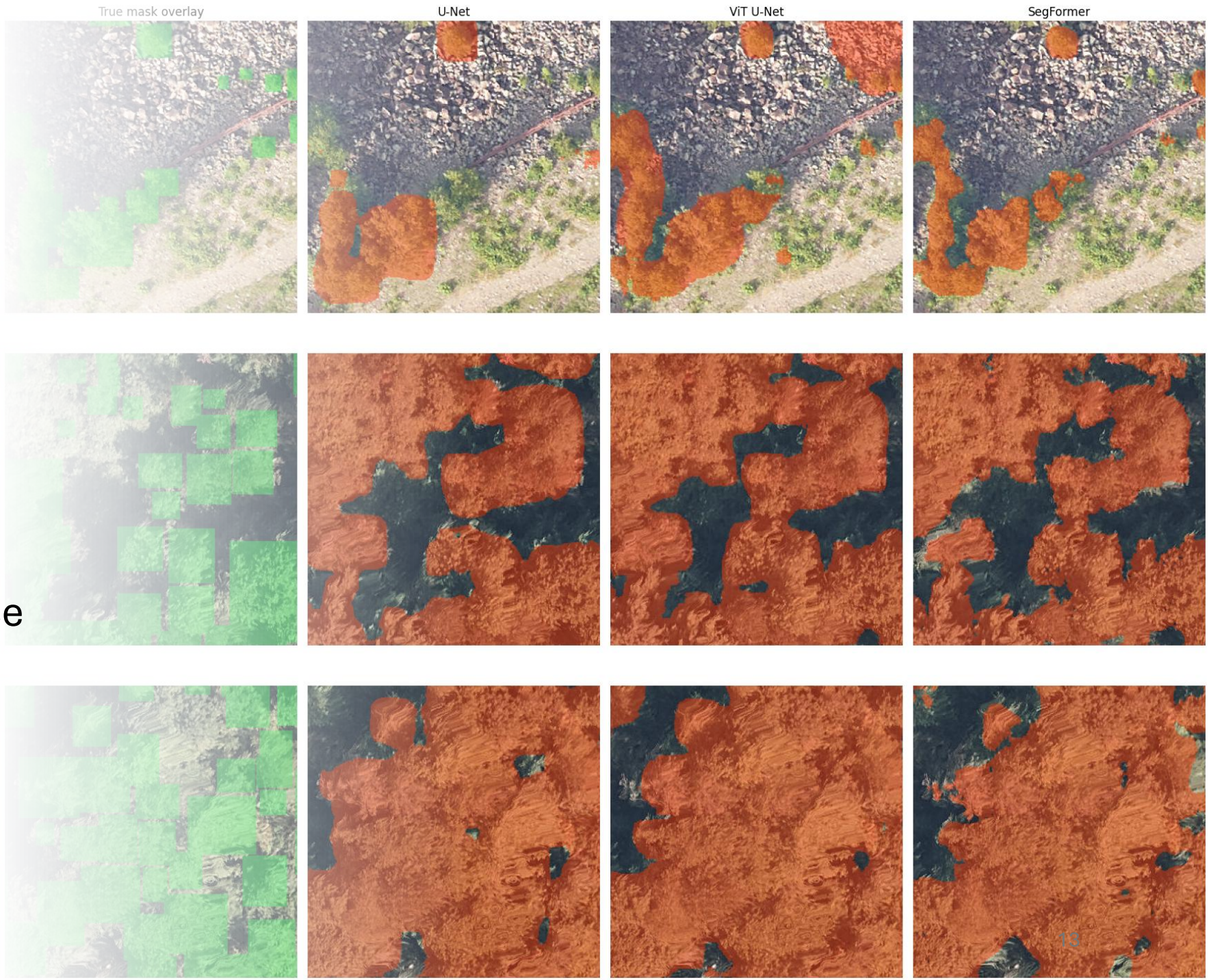
Mask opacity

0.30

Center: 37.09315 N, -119.78202 E | CRS: EPSG:32611 | Bounds: [37.0930, -119.7823] → [37.0933, -119.7818]

Scene Metrics

Tree count	Tree density (trees/ha)	Spatial distribution
6	37.50	uniform
Avg crown area (m^2)	**Mean NN distance (m)**	**Fragmentation**
91.02	16.21	0.010986
Crown variance (m^{22})	**NN index**	**Canopy area (m^2)**
16882.26	1.98	546.13



Thanks!
Questions

Github:

Data source